

Model Comparison — Holdout 2022–2025 [RMSE loss variant]
RMSE, MAE, Skill vs RW, R², Hit Rate | DM = Harvey et al. (1997) vs RW | CW = Clark-West (2007) vs OLS

Horizon	CatBoost RMSE	CatBoost MAE	CatBoost Skill%	CatBoost R²	CatBoost Hit	CatBoost p(DM)	CatBoost p(CW)	HTBoost RMSE	HTBoost MAE	HTBoost Skill%	HTBoost R²	HTBoost Hit	HTBoost p(DM)	HTBoost p(CW)	Lasso RMSE	Lasso MAE	Lasso Skill%	Lasso R²	Lasso Hit	Lasso p(DM)	Lasso p(CW)	SARIMA RMSE	SARIMA MAE	SARIMA Skill%	SARIMA R²	SARIMA Hit	SARIMA p(DM)	SARIMA p(CW)	OLS RMSE	OLS MAE	OLS Skill%	OLS R²	OLS Hit	OLS p(DM)	OLS p(CW)
0w	0.0650	0.0503	+9.0%	0.171	61.3%	<.001	—	0.0601	0.0472	+15.9%	0.292	68.5%	<.001	—	0.0632	0.0490	+11.5%	0.217	65.8%	<.001	—	0.0673	0.0507	+5.7%	0.110	61.7%	<.001	—	—	—	—	—	—	—	
1w	0.1008	0.0773	+11.8%	0.221	67.0%	<.001	—	0.0995	0.0768	+12.9%	0.240	64.3%	<.001	—	0.0994	0.0758	+12.9%	0.241	66.1%	<.001	—	0.1105	0.0845	+3.2%	0.063	57.5%	0.032	—	—	—	—	—	—	—	
2w	0.1203	0.0937	+12.5%	0.234	66.8%	<.001	—	0.1157	0.0897	+15.9%	0.291	66.8%	<.001	—	0.1182	0.0914	+14.0%	0.260	67.3%	<.001	—	0.1322	0.1031	+3.8%	0.074	65.9%	0.007	—	—	—	—	—	—	—	
1m	0.1395	0.1080	+17.4%	0.317	70.6%	<.001	0.018	0.1412	0.1051	+16.5%	0.301	72.5%	<.001	0.135	0.1416	0.1053	+16.2%	0.297	71.6%	<.001	0.452	0.1614	0.1251	+4.5%	0.086	61.0%	<.001	—	0.1399	0.1058	+17.2%	0.313	67.4%	<.001	—
3m	0.2567	0.2088	+15.2%	0.281	72.7%	<.001	0.381	0.2348	0.1886	+22.5%	0.399	76.1%	<.001	0.260	0.2301	0.1865	+24.0%	0.423	75.6%	<.001	0.503	0.2750	0.2340	+9.2%	0.175	77.5%	<.001	—	0.2179	0.1723	+28.1%	0.482	75.6%	0.004	—
6m	0.2897	0.2430	+20.2%	0.363	74.5%	0.008	0.254	0.2400	0.1984	+33.9%	0.563	87.2%	<.001	0.200	0.2339	0.1895	+35.6%	0.585	85.7%	<.001	0.288	0.3187	0.2764	+12.2%	0.229	82.7%	<.001	—	0.2232	0.1604	+38.5%	0.622	87.8%	0.009	—
12m	0.1916	0.1518	+10.3%	0.189	70.0%	0.092	<.001	0.2060	0.1600	+3.6%	0.063	64.1%	0.313	<.001	0.2210	0.1677	-3.4%	-0.079	51.2%	0.661	0.004	0.2080	0.1631	+2.6%	0.044	53.5%	0.281	—	0.2590	0.2124	-21.2%	-0.481	46.5%	0.986	—